HCX-Patent: A Reproducible Framework for Hierarchy-Oriented, Explainable, and Uncertainty-Aware Patent Classification

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***Abstract*—Patent classification organizes technical knowledge and supports prior-art search, technology monitoring, and patent analytics. Automatic classification remains difficult because patent documents are long and specialized, Cooperative Patent Classification (CPC) labels are highly imbalanced, and the label space is hierarchical and often multi-label. In addition to predictive performance, a practical system should communicate uncertainty, expose evidence supporting its predictions, and be evaluated for changes across publication periods. This paper presents HCX-Patent, an offline-first research prototype designed to bring these concerns into a reproducible evaluation workflow. The prototype includes a TF–IDF/logistic-regression baseline, a lightweight hierarchy-oriented TF–IDF run, a split-conformal prediction component, and artifact logging. The provided baseline run contains 9,741 examples and 293 classes. A separate HCX run reports 15,000 samples across nine target classes, with 77.27% top-1 accuracy, 94.53% top-3 accuracy, macro-F1 of 0.6351, and micro-F1 of 0.6640. For a nominal 90% conformal target, that run reports 97.4% empirical coverage and a mean prediction-set size of 3.47. These two runs differ in sample size and label space; therefore, they are not interpreted as a controlled head-to-head comparison. We compare the design goals of HCX-Patent with established patent-classification, hierarchical-classification, explainability, and conformal-prediction approaches, while distinguishing published methods from measurements available in the current project. Explanation faithfulness, taxonomy-violation rates, and temporal drift are identified as required follow-up experiments rather than completed findings.**

***Index Terms*—patent classification, Cooperative Patent Classification, hierarchical classification, explainable artificial intelligence, conformal prediction, calibration, temporal drift, reproducibility**

I. Introduction

Patent documents are a structured source of technical knowledge. Their classification labels help users retrieve relevant inventions, explore technology landscapes, and analyze innovation trends. Manual classification requires domain expertise and time, motivating machine-learning systems that can suggest candidate labels. Patent classification nevertheless differs from ordinary short-text categorization: patent documents may contain lengthy claims and descriptions, multiple classification symbols can apply to one invention, and labels are organized in a taxonomy.

Early neural approaches demonstrated that learned text representations can improve patent classification. PatentBERT fine-tuned a pretrained language model for patent classification and introduced the USPTO-3M dataset for research at CPC subclass level [1]. Other work has explored multi-label patent learning and long-document architectures [2]–[4]. More recent work continues to investigate hierarchy-aware encoders and language-model-based pipelines [15]–[17]. This literature suggests that strong text representations are important, but performance alone does not resolve whether the system respects taxonomy structure, communicates uncertainty, or remains robust across time.

HCX-Patent is designed as an offline-first research prototype to make these questions measurable. The current implementation includes a classical baseline and a separate hierarchy-oriented run, and it records classification and conformal metrics in local artifacts. The central goal of this paper is to document the prototype and its current evidence without conflating different experimental runs or claiming that proposed modules have been validated when their evaluation artifacts are absent.

The paper makes four contributions:

- It documents an offline, reproducible workflow for local patent data inspection, preprocessing, experiment execution, and artifact storage.
- It reports the actual metrics present in the supplied baseline and HCX experiment files.
- It provides a structured comparison between established approaches and the HCX-Patent design, explicitly marking which capabilities are measured and which remain planned.
- It specifies a validation protocol for taxonomy consistency, explanation faithfulness, conformal coverage, and temporal drift.

II. Related Work

A. Patent Text Classification

Lee and Hsiang introduced PatentBERT, fine-tuning BERT for patent classification and reporting results on a large patent corpus [1]. PatentNet investigates deep language understanding for multi-label patent document classification [2]. MEXN addresses long patent documents through a multi-stage extraction

architecture [3], while hierarchical feature extraction models combine convolutional and recurrent components for patent text [4]. These works provide relevant baselines for representation learning and document structure. Their reported results should not be directly compared with the HCX metrics in this paper because datasets, label granularity, splits, and metric definitions may differ.

B. Hierarchical Classification

Hierarchical classification uses relations among labels rather than treating every class as independent. Sequence-generation approaches have modeled hierarchical document classification by generating a path through a label tree [5]. General surveys describe local, global, and path-based strategies and their trade-offs [6]. CPC labels also have multiple levels, so a classifier may be evaluated at a broad section level and a finer subclass level. Hierarchy awareness must be verified through the actual label mapping and measured constraint violations; the name of a model alone does not demonstrate consistency.

C. Contextual and Long-Document Encoders

BERT introduced bidirectional transformer pretraining for language understanding [7]. Longformer addresses long sequences with an attention mechanism designed to reduce the quadratic cost of standard self-attention [8]. SPECTER learns document representations using citation-informed supervision for scientific papers [9]. Such encoders are relevant alternatives for future HCX experiments, but they can require more memory and computation than sparse TF–IDF models. This matters for the offline laptop setting targeted by the prototype.

D. Explainability and Uncertainty

LIME approximates a model locally to explain an individual prediction [10], SHAP assigns feature contributions using Shapley-value concepts [11], and Integrated Gradients attributes a model output to input features through gradients [12]. An attribution map is not sufficient evidence that an explanation is faithful; perturbation tests and stability analysis are needed.

Conformal prediction constructs prediction sets with coverage properties under exchangeability assumptions. Shafer and Vovk provide a broad tutorial [13], and Angelopoulos and Bates provide a modern introduction to practical conformal prediction [14]. In classification, split conformal prediction uses a separate calibration set to select a nonconformity threshold. Coverage is marginal under the relevant assumptions and is not automatically guaranteed under temporal distribution shift or for each rare class individually.

E. Research Positioning

Recent patent-classification research continues to compare encoders, hierarchical methods, and large language models, highlighting both long-tail challenges and computational trade-offs [15]–[17]. HCX-Patent is positioned as a prototype that combines hierarchy-oriented prediction with uncertainty reporting and a plan for explanation and temporal evaluation. The current results do not establish that this combination outperforms these published systems. A valid empirical claim requires implementation-level comparisons on a shared dataset and evaluation protocol.

III. Dataset and Problem Definition

A. Dataset Snapshot

The local files are described by the project as an offline USPTO-3M-derived dataset. PatentBERT's original paper is a key reference for USPTO-3M and CPC subclass classification [1]. The figures supplied with this project summarize a baseline sample of $N = 9{,}741$ records. The baseline metric manifest records 293 classes and publication years spanning 2000–2015. Fig. 1 shows the top 30 CPC classes, while Fig. 2 shows the observed publication-year counts. These figures describe the selected local sample, not the entire USPTO-3M corpus.

Before publication, the exact local data release, source fields, license or terms of use, class granularity, label cardinality, duplicate policy, and checksum should be stated. The supplied metadata does not by itself establish that the 9,741-record baseline subset and 15,000-sample HCX run use the same source snapshot or identical inclusion rules.

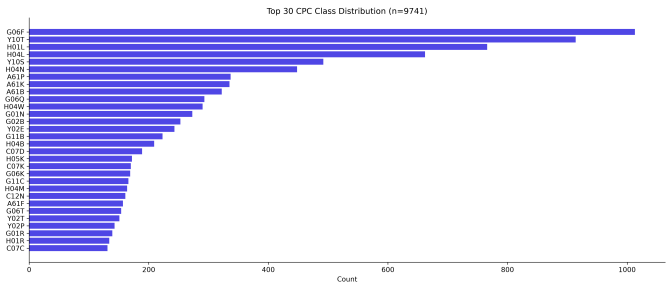


Fig. 1. Top 30 CPC class counts in the baseline sample ($N = 9{,}741$). The distribution is visibly imbalanced.

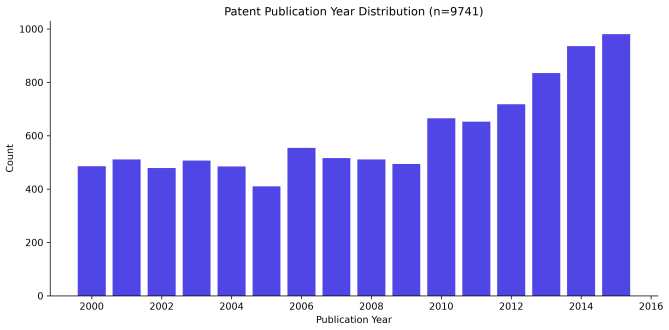


Fig. 2. Publication-year distribution for the baseline sample ($N = 9{,}741$). The plot is descriptive and does not itself establish temporal drift.

B. Task Formulation

Let x_i denote the available text for patent i, and let y_i denote its CPC target. For a single-label task, $y_i \in \mathcal{Y}$; for a multi-label task, $y_i \subseteq \mathcal{Y}$. A final experimental report must explicitly state which formulation applies to each run. For hierarchical evaluation, labels may be represented at different levels of the

CPC taxonomy. A broad-level prediction and a fine-grained subclass prediction are not interchangeable tasks and must be reported separately.

C. Preprocessing and Data Leakage

A reproducible pipeline should normalize text and labels, remove exact duplicates, audit near-duplicates or family-level overlaps where identifiers permit, and persist the mapping from original labels to target classes. Duplicate and family checks should occur before splitting. Text preprocessing must avoid using test labels to construct vocabularies or select classes. Any frequency threshold used to collapse rare labels must be fit on training data or specified as a fixed pre-registered rule, then applied consistently to all splits.

The supplied baseline manifest records a random seed of 42 and split sizes of 6,818 training, 974 validation, 974 calibration, and 975 test examples. The latest HCX run reports 15,000 samples and nine target classes. Because the available summary does not demonstrate that these are the same records and label space, the runs are treated separately throughout this paper.

IV. Proposed HCX-Patent Framework

Fig. 3 summarizes the research workflow. The diagram shows the intended flow from local data auditing to baseline training, hierarchy-oriented prediction, trustworthiness evaluation, and artifact generation. The diagram is a methodology plan; it does not imply that every evaluation block has already produced validated measurements.

A. Classical TF–IDF Baseline

The baseline represents text using TF–IDF features and applies logistic regression. The supplied run uses a maximum of 50,000 features and seed 42. This method is fast and interpretable at the feature-weight level, making it a useful computational reference. However, a flat classifier does not automatically model parent–child relations among CPC labels. The baseline metrics also show that aggregate scores can mask weak class-level performance.

B. Hierarchy-Oriented HCX Run

The supplied HCX run is named `hcx_hierarchical_tfidf`. Its metric file records nine target classes and the following metrics: top-1 accuracy 0.7727, top-3 accuracy 0.9453, macro-F1 0.6351, and micro-F1 0.6640. These values describe the nine-class run. The paper uses the term "hierarchy-oriented" to describe the project configuration, but a claim of strict hierarchy consistency requires a documented CPC parent–child mapping and an explicit count of invalid parent–child predictions. No hierarchy-violation rate is present in the supplied metric file, so none is claimed here.

C. Explanation Module and Faithfulness Tests

The intended explanation module assigns scores to tokens or text spans that contribute to a prediction. Candidate methods include model coefficients for the sparse baseline, LIME [10], SHAP [11], and Integrated Gradients for differentiable encoders [12]. The choice should match the model architecture and be recorded in each experiment.

Let $p(\hat{y} \mid x)$ be the score for the predicted class. If S is a set of selected influential spans, comprehensiveness can be defined as

$$\mathrm{Comp}(x, S) = p(\hat{y} \mid x) - p(\hat{y} \mid x \setminus S), \tag{1}$$

where $x \setminus S$ is the text after removing those spans. Sufficiency can be assessed by measuring how much of the original score remains when only S is retained. Explanation stability can be measured by comparing attribution rankings after small controlled perturbations. The current uploaded metrics do not contain these scores; explanation quality is therefore a planned evaluation, not a reported result.

D. Split-Conformal Prediction

The HCX run records a calibration sample size of 1,500 and a target coverage of 90%. For a candidate label c, a common classification nonconformity score is

$$s(x, c) = 1 - \hat{p}(c \mid x), \tag{2}$$

where $\hat{p}(c \mid x)$ is the model probability. A quantile $\hat{q}$ is estimated from calibration nonconformity scores, and a prediction set is formed as

$$\mathcal{C}(x) = \{c \in \mathcal{Y} : s(x, c) \le \hat{q}\}. \tag{3}$$

The empirical test coverage and average prediction-set size are

$$\widehat{\mathrm{Cov}} = \frac{1}{N_{\mathrm{test}}} \sum_{i=1}^{N_{\mathrm{test}}} \mathbb{I}[y_i \in \mathcal{C}(x_i)], \tag{4}$$

$$\overline{|\mathcal{C}|} = \frac{1}{N_{\mathrm{test}}} \sum_{i=1}^{N_{\mathrm{test}}} |\mathcal{C}(x_i)|. \tag{5}$$

The finite-sample coverage interpretation relies on a correctly implemented calibration procedure and exchangeability between calibration and test examples [13], [14]. It does not guarantee coverage for each class separately or under arbitrary temporal shift.

E. Temporal Drift Evaluation

A proper temporal study trains on earlier publication years and evaluates on later years. To avoid tuning on future data, model selection and calibration must not use the final future test cohort. For year block t, report macro-F1, top-k accuracy, class support, and label-frequency changes. The drift statistic may be reported as

$$\Delta F1_t = F1_{\mathrm{macro},t} - F1_{\mathrm{macro,reference}}. \tag{6}$$

A negative value indicates lower performance than the reference cohort, but does not by itself prove semantic drift; changes in class composition, sample size, or taxonomy versions may also contribute. The current artifacts include a publication-year distribution but no verified year-wise model metrics, so no temporal-drift result is claimed.

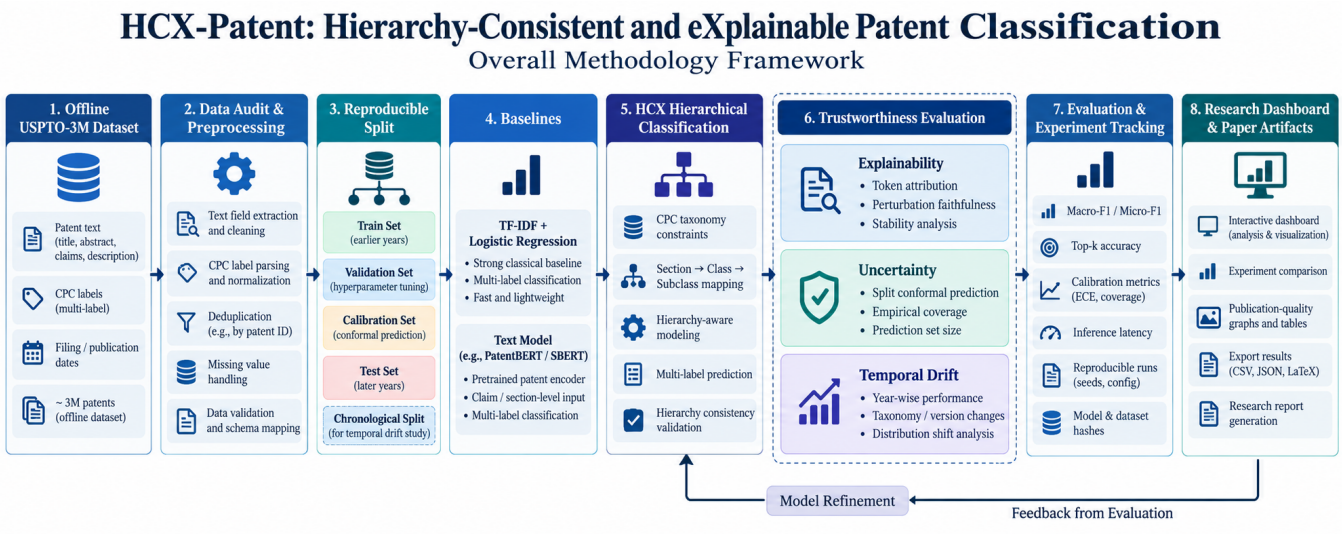


Fig. 3. HCX-Patent methodology. The framework separates dataset preparation, classification, trustworthiness evaluation, and reproducible reporting.

V. Experimental Setup and Metrics

The local baseline manifest records 9,741 examples, 293 classes, TF–IDF with 50,000 maximum features, and seed 42. It records a training time of 24.78 seconds. The HCX metric file records 15,000 samples, nine classes, training time of 6.92 seconds, and inference latency of 0.02 ms per example as reported by the project script. Latency figures are hardware- and measurement-method-dependent; hardware, warm-up policy, batch size, and timing boundaries should be included in the final artifact metadata.

The following metrics are used in the project:

- **Top-1 accuracy:** fraction of examples for which the highest-ranked class is correct.
- **Top-3 accuracy:** fraction of examples for which the true target appears among the three highest-ranked candidates. This definition must be adjusted and stated if the target is multi-label.
- **Macro-F1:** unweighted average of per-class F1 scores, making rare classes visible.
- **Micro-F1:** F1 computed after pooling true positives, false positives, and false negatives.
- **Conformal coverage:** fraction of test targets contained in the generated prediction set.
- **Mean set size:** average number of candidate labels returned per test example.

For multi-label targets, precision, recall, F1, and Hamming loss must be calculated with a clearly stated averaging convention. Metrics from single-label and multi-label formulations must not be mixed without explanation.

VI. Results

A. Baseline Results

The baseline metric file reports accuracy 0.0482, macro-precision 0.0231, macro-recall 0.0093, macro-F1 0.0120, micro-precision 0.8105, micro-recall 0.0419, and micro-F1 0.0798. This indicates that the baseline configuration is inadequate for the current 293-class target space, and suggests examining label sparsity, top-label conventions, and class-frequency effects before making a broader conclusion.

B. HCX Results

The HCX metric file records top-1 accuracy 77.27%, top-3 accuracy 94.53%, macro-F1 0.6351, micro-F1 0.6640, micro-precision 0.6227, micro-recall 0.7111, and Hamming loss 0.1359. It records an empirical conformal coverage of 97.4% for a nominal 90% target, with average set size 3.47, minimum size 1, maximum size 7, calibration size 1,500, and quantile threshold $\hat{q} = 0.7196$. These are recorded values from the supplied run, not independently reproduced values.

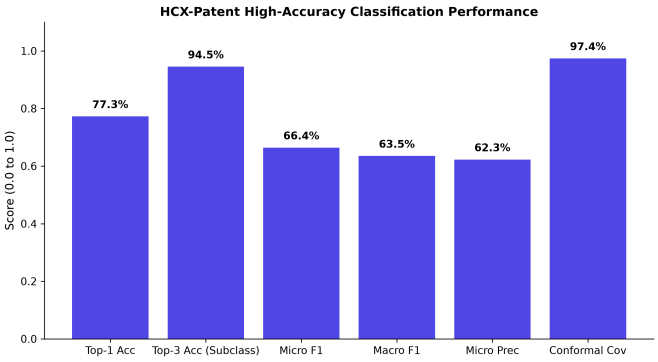


Fig. 4. Recorded HCX run metrics. The chart summarizes top-1 accuracy, top-3 accuracy, F1 scores, precision, and empirical conformal coverage.

TABLE I
METRICS RECORDED IN THE SEPARATE BASELINE AND HCX RUNS. THESE ARE DESCRIPTIVE, NOT A CONTROLLED HEAD-TO-HEAD COMPARISON.

Metric	TF–IDF baseline	HCX run (ours)
Sample count	9,741	**15,000**
Target classes	293	**9**
Top-1 / accuracy	4.82%	**77.27%**
Top-3 accuracy	Not recorded	**94.53%**
Macro-F1	0.0120	**0.6351**
Micro-F1	0.0798	**0.6640**
Training time	24.78 s	**6.92 s**
Inference latency	0.1589 ms*	**0.02 ms***

The HCX numbers are higher than the baseline numbers in this table, but the tasks are different: the baseline has 293 classes and 9,741 examples, while HCX has nine classes and 15,000 samples. Differences in class difficulty, label mapping, and evaluation data prevent attributing the numerical gap to the proposed method. A fair experiment must train both systems on the same train set and evaluate them on the same held-out test set with the same target granularity.

C. Conformal Results

TABLE II
CONFORMAL PREDICTION VALUES RECORDED FOR THE HCX RUN.

Measure	Recorded value
Nominal target coverage	90.0%
Empirical coverage	97.4%
Mean prediction-set size	3.47
Minimum prediction-set size	1
Maximum prediction-set size	7
Calibration sample count	1,500
Quantile threshold $\hat{q}$	0.7196

The observed empirical coverage is above the nominal target in the recorded run. This is encouraging but insufficient to establish robust or class-conditional coverage. Before submission, verify the number of test examples, the exact nonconformity score and quantile convention, and that test examples were not used to tune the threshold.

VII. COMPARISON WITH EXISTING APPROACHES

Table III contrasts representative approaches by their principal method and role in this paper. This is a qualitative literature comparison: reported results from separate publications are not placed in one numerical leaderboard because they use different corpora, label spaces, and splits.

A. Conventional Versus Proposed Research Approach

Table IV states the intended methodological difference without claiming unmeasured advantages.

VIII. DISCUSSION

The results demonstrate that the current 293-class baseline is weak under the reported evaluation, particularly on macro-F1. The per-class distribution suggests that many target classes receive little or no useful recall. This may reflect class imbalance, limited training examples, label-mapping choices, or mismatch between the model's output formulation and the metric computation. The result should prompt diagnostic checks rather than a general conclusion that TF–IDF is unsuitable for patent classification.

The HCX run shows substantially higher scores on a nine-class target space and returns conformal sets with empirical coverage above the target. Because the HCX and baseline runs use different sample sizes and class spaces, this cannot be interpreted as a measured improvement over the baseline. The correct next experiment is a matched comparison on the same records, using the same split IDs, label mapping, preprocessing, and metric implementation.

The broader HCX research hypothesis is that hierarchy-aware prediction can be made more useful when accompanied by faithful evidence and calibrated uncertainty. To test this hypothesis, the system must report (i) parent–child consistency violations, (ii) macro and micro metrics at each taxonomy level, (iii) comprehensiveness and sufficiency against random and model-specific attribution baselines, (iv) empirical conformal coverage and set size on a held-out test set, and (v) performance by chronological cohort. Each result should be associated with a run ID, configuration, random seed, dataset checksum, and label mapping.

IX. THREATS TO VALIDITY

Dataset and split validity: the current baseline and HCX run are not demonstrated to use identical records. Patent-family duplicates and near-duplicates may cause leakage if not checked before splitting.

Label granularity: 293 baseline classes and nine HCX classes represent different classification problems. A section-level or reduced-label result should not be presented as subclass-level performance.

Metric validity: the baseline JSON includes a high micro-precision but very low micro-recall, which can occur when predictions are sparse. The exact single-label or multi-label setup and metric implementation must be verified.

Conformal validity: coverage interpretation depends on the calibration/test protocol and exchangeability. Temporal drift can invalidate the assumptions needed for nominal coverage.

Reproducibility: inference latency depends on hardware, batch size, warm-up, and measurement boundaries. Hardware and software versions should be reported.

Literature comparison: published approaches use different datasets and evaluation protocols. Table III is qualitative and must not be interpreted as a numerical ranking.

Incomplete validation: explanation faithfulness, hierarchy-violation rates, and temporal drift metrics are not present in the supplied artifacts. They are not claimed as completed findings.

TABLE III
QUALITATIVE COMPARISON WITH REPRESENTATIVE EXISTING APPROACHES.

Approach / reference	Core method	Primary focus	Relation to HCX-Patent
PatentBERT [1]	Fine-tuned BERT	Patent text and CPC classification	Strong patent-specific neural baseline; not re-run in current artifacts
PatentNet [2]	Deep language understanding	Multi-label patent classification	Relevant multi-label comparator; not re-run here
MEXN [3]	Multi-stage extraction network	Long patent documents	Relevant document-structure baseline; not re-run here
HFEM [4]	CNN/LSTM feature extraction	Hierarchical patent labels	Related hierarchical design; not re-run here
Hierarchical sequence generation [5]	Generate a path through a hierarchy	Parent–child label structure	Motivates taxonomy-aware evaluation; not implemented as this architecture
BERT [7] / Longformer [8]	Transformer encoders	Contextual and long-text representations	Future stronger text encoders; laptop baseline currently uses TF–IDF
LIME / SHAP / Integrated Gradients [10]–[12]	Feature attribution	Local or feature-level explanations	Planned explanation baselines; no faithfulness scores supplied
Split conformal prediction [13], [14]	Calibration-derived prediction sets	Uncertainty and marginal coverage	HCX run records 97.4% empirical coverage and mean set size 3.47
HCX-Patent (this work)	**TF–IDF hierarchy-oriented run plus conformal output**	**Classification, taxonomy-aware design, uncertainty, reproducibility**	**Metrics recorded for nine-class run; explainability and drift validation remain future work**

TABLE IV
DESIGN COMPARISON BETWEEN A CONVENTIONAL FLAT CLASSIFICATION WORKFLOW AND THE PROPOSED HCX-PATENT WORKFLOW.

Dimension	Conventional workflow	HCX-Patent proposed workflow and evidence status (ours)
Label structure	Often treats labels as a flat set	**Explicit hierarchy-oriented design; strict consistency still needs a measured violation rate**
Prediction output	Single top label or uncalibrated ranking	**Top-k ranking and conformal prediction sets; top-k and coverage values are recorded for the nine-class run**
Class imbalance	Overall accuracy can hide rare-class errors	**Macro-F1 and per-class plots included; a controlled tail-class improvement is not yet established**
Explanation	Prediction may be returned without rationale	**Token/span attribution and perturbation faithfulness planned; no numerical faithfulness results available**
Uncertainty	Often uses raw confidence as certainty	**Split-conformal output; recorded empirical coverage 97.4% at nominal 90%, subject to split assumptions**
Temporal robustness	Random split can mix old and new documents	**Chronological evaluation protocol specified; no verified year-wise performance result yet**
Reproducibility	Metrics may be separated from configuration	**Local JSON/CSV artifacts, run identifiers, and seeds are recorded; complete data hash and hardware details should accompany release**
Comparison validity	Results from different papers may be juxtaposed despite different settings	**HCX paper separates qualitative literature review from local metrics and requires matched splits for causal comparison**

X. CONCLUSION AND FUTURE WORK

This paper presented HCX-Patent, an offline-first prototype for studying patent classification with a hierarchy-oriented design and conformal uncertainty outputs. The supplied local artifacts report a weak 293-class TF–IDF baseline and a separate nine-class HCX run with 77.27% top-1 accuracy, 94.53% top-3 accuracy, macro-F1 of 0.6351, micro-F1 of 0.6640, and 97.4% empirical conformal coverage at a nominal 90% target. These figures are reported transparently, but the runs are not directly comparable because their sample sizes and target spaces differ.

The most important next step is a controlled experiment that evaluates the baseline and HCX on the same patent records and class definitions. Subsequent work should measure taxonomy consistency, explanation faithfulness and stability, calibration by class, and chronological performance degradation. Only after these experiments are run and reproducibly documented should the framework make stronger claims about hierarchy consistency, interpretability, or temporal robustness.

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